Solution to Mino Puzzle

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Part I

1. **Answer:** Such an arrangement is not possible.

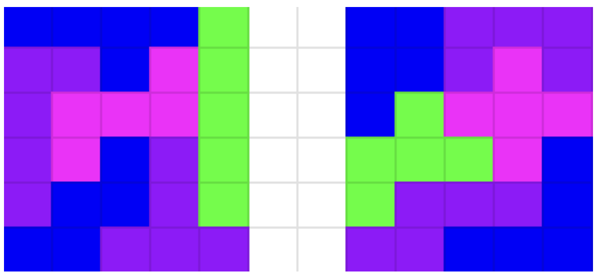
   Solution: This question introduces the idea of **parity**, which is a topic of interest in both stacking mechanics and identifying **perfect clears**.
   Consider the rectangle described in the problem. Shade each of the 28 cells alternating black or white. Because the number of cells within the rectangle is even, there will be 14 black and 14 white squares no matter the coloration.
   However, coloring each tetromino with the same pattern, it appears that there are always 2 black and 2 white cells except for the T piece. Depending on the coloration, it will have 3 black cells or 3 white cells, with only 1 of the alternative cell color. Since the parities of the 7 pieces together and the rectangle are not equal, such an arrangement is not possible.

2. **Answer:** An even number of spaces away.

   Solution: Though the cleanest solution would be to place one T-piece on top of another, as this will most easily result in an even number of black and white squares, placing the second T-piece anywhere an even number of pieces away will make them result in the same parity, hence resulting in an even parity.

3. **Answer:**

4. **Answer:**



Part II

1. **Answer/Solution:** Despite many efforts, there is no general formula for an n-mino, though there are estimates to an upper and lower bound for a relation between n-minos with different rotational properties.